The Abstraction Gap Between Mathematics and Physics: A Geometric Perspective on Dimensional Analysis

Fredrik Bierich
fredrik.bierich@gmail.com

November 12, 2026

Abstract

Designed as a seed for a mathematical corpus, this research artifact attempts to bridge the abstraction gap between mathematics and physics by redefining the arcus functions and the circular functions from the geometric quantities of a circular sector. The utility of this framework is then demonstrated by modeling simple harmonic motion and uniform circular motion, providing an exploratory foundation for evaluating formal reasoning and improving dimensional consistency in computational models.

0.1 Circular Sector

0.1.1 Arc Length

$$r \in \mathbb{R}^+ \tag{1}$$

$$x \in [-r, r] \tag{2}$$

$$r \mapsto \overline{OA} \tag{3}$$

$$x \mapsto \overline{OB_0} \tag{4}$$

$$r \mapsto \overline{OC_n}, n \geq 0 \tag{5}$$

$$a_n \mapsto \overline{OB_n}, n \geq 0 \tag{6}$$

$$s_n \mapsto \overline{AC_n}, n \geq 0 \tag{7}$$

$$s \mapsto \operatorname{arc} AC_0 \tag{8}$$

$$s = \lim_{n \to \infty} 2^n s_n \tag{9a}$$

$$s_n = \sqrt{r^2 + r^2 - 2ra_n} \tag{9b}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \tag{9c}$$

$$a_0 = x, x \neq r \tag{9d}$$

$$s = \lim_{n \to \infty} 2^n s_n \tag{10a}$$

$$s_n = \sqrt{2r^2 - 2ra_n} \tag{10b}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \tag{10c}$$

$$a_0 = x, x \neq r \tag{10d}$$

$$s = \lim_{n \to \infty} 2^n s_n \tag{11a}$$

$$s_n = \sqrt{2r^2 - 2ra_n} \tag{11b}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{11c}$$

$$a_0 = x, x \neq r \tag{11d}$$

$$s = \lim_{n \to \infty} 2^n s_n \tag{12a}$$

$$s_n = \sqrt{2r^2 - 2ra_n} \tag{12b}$$

$$a_n = r \cdot \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{12c}$$

$$a_0 = x, x \neq r \tag{12d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{13a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{13b}$$
$$a_n = r\sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{13c}$$
$$a_0 = x, x \neq r \tag{13d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{14a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{14b}$$
$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{14c}$$
$$a_0 = x, x \neq r \tag{14d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{15a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{15b}$$
$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}}, n \geq 1 \tag{15c}$$
$$a_0 = x, x \neq r \tag{15d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{16a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{16b}$$
$$a_n = r\sqrt{\frac{r + a_{n-1}}{2r}}, n \geq 1 \tag{16c}$$
$$a_0 = x \tag{16d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{17a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{17b}$$
$$a_n = r\sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}}, n \geq 1 \tag{17c}$$
$$a_0 = x \tag{17d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{18a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{18b}$$
$$a_n = r\sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}}, n \geq 1 \tag{18c}$$
$$a_0 = x \tag{18d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{19a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{19b}$$
$$a_n = r \cdot \frac{\sqrt{2r^2 + 2ra_{n-1}}}{\sqrt{(2r)^2}}, n \geq 1 \tag{19c}$$
$$a_0 = x \tag{19d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{20a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{20b}$$
$$a_n = r \cdot \frac{\sqrt{2r^2 + 2ra_{n-1}}}{|2r|}, n \geq 1 \tag{20c}$$
$$a_0 = x \tag{20d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{21a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{21b}$$
$$a_n = r \cdot \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \tag{21c}$$
$$a_0 = x \tag{21d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{22a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{22b}$$
$$a_n = \frac{r\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \tag{22c}$$
$$a_0 = x \tag{22d}$$

$$s = \lim_{n\to\infty} 2^n s_n \tag{23a}$$
$$s_n = \sqrt{2r^2 - 2ra_n} \tag{23b}$$
$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{23c}$$
$$a_0 = x \tag{23d}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{24a}$$
$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{24b}$$
$$a_0 = x \tag{24c}$$

0.1.1.1 Quadrant

$$Q_r \stackrel{\text{def}}{=} \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{25a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{25b}$$

$$a_0 = 0 \tag{25c}$$

0.1.2 Area

$$r \in \mathbb{R}^+ \tag{26}$$

$$s \in (0, 4Q_r] \tag{27}$$

$$r \mapsto \overline{OA} \tag{28}$$

$$s \mapsto \operatorname{arc} AC_0 \tag{29}$$

$$A \mapsto \operatorname{area} \triangledown OAC_0 \tag{30}$$

$$A = \frac{rs}{2} \tag{31}$$

0.1.3 Arcus Functions

0.1.3.1 Arccosine

$$\begin{aligned} \arccos\colon \{(r, x) \in \mathbb{R}^+ \times \mathbb{R} \mid -r \leq x \leq r\} &\to \mathbb{R} \\ (r, x) &\mapsto s \in [0, 2Q_r] \end{aligned} \tag{32}$$

$$\arccos(r, x) = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{33a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{33b}$$

$$a_0 = x \tag{33c}$$

0.1.3.2 Arcsine

$$\begin{aligned} \arcsin\colon \{(r, y) \in \mathbb{R}^+ \times \mathbb{R} \mid -r \leq y \leq r\} &\to \mathbb{R} \\ (r, y) &\mapsto s \in [-Q_r, Q_r] \end{aligned} \tag{34}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y)}{2} \tag{35}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2} \tag{36}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2} \tag{37}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2} \tag{38}$$

$$\arcsin(r, y) = \frac{\arccos(r, -y) - \arccos(r, y)}{2} \tag{39}$$

0.1.3.2.1 Negation

$$s = \arcsin(r, -y) \tag{40}$$
$$\sin(r, s) = -y \tag{41}$$
$$-\sin(r, s) = -(-y) \tag{42}$$
$$\sin(r, -s) = -(-y) \tag{43}$$
$$\sin(r, -s) = y \tag{44}$$
$$-s = \arcsin(r, y) \tag{45}$$
$$-(-s) = -\arcsin(r, y) \tag{46}$$
$$s = -\arcsin(r, y) \tag{47}$$
$$\arcsin(r, -y) = -\arcsin(r, y) \tag{48}$$

0.1.4 Circular Functions

0.1.4.1 Cosine

$$\begin{aligned} \cos\colon \mathbb{R}^+ \times \mathbb{R} &\to \mathbb{R} \\ (r, s) &\mapsto x \in [-r, r] \end{aligned} \tag{49}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{50a}$$
$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{50b}$$
$$a_0 = x \tag{50c}$$

$$m \in \mathbb{N}_0 \tag{51a}$$
$$s \approx 2^m \sqrt{2r^2 - 2rb_{m,m}} \tag{51b}$$
$$b_{m,n} = \frac{\sqrt{2r^2 + 2rb_{m,n-1}}}{2}, 1 \leq n \leq m \tag{51c}$$
$$b_{m,0} = x \tag{51d}$$

$$m \in \mathbb{N}_0 \tag{52a}$$
$$x \approx b_{m,m} \tag{52b}$$
$$b_{m,n-1} = \frac{\sqrt{2r^2 + 2rb_{m,n}}}{2}, 1 \leq n \leq m \tag{52c}$$
$$2^m \sqrt{2r^2 - 2rb_{m,0}} = s \tag{52d}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{53a}$$
$$b_{m,n-1} = \frac{\sqrt{2r^2 + 2rb_{m,n}}}{2}, 1 \leq n \leq m \tag{53b}$$
$$2^m \sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{53c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{54a}$$
$$2b_{m,n-1} = 2 \cdot \frac{\sqrt{2r^2 + 2rb_{m,n}}}{2}, 1 \leq n \leq m \tag{54b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{54c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{55a}$$
$$2b_{m,n-1} = \frac{2\sqrt{2r^2 + 2rb_{m,n}}}{2}, 1 \leq n \leq m \tag{55b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{55c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{56a}$$
$$2b_{m,n-1} = \sqrt{2r^2 + 2rb_{m,n}}, 1 \leq n \leq m \tag{56b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{56c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{57a}$$
$$(2b_{m,n-1})^2 = \left(\sqrt{2r^2 + 2rb_{m,n}}\right)^2, 1 \leq n \leq m \tag{57b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{57c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{58a}$$
$$2^2b_{m,n-1}^2 = \left(\sqrt{2r^2 + 2rb_{m,n}}\right)^2, 1 \leq n \leq m \tag{58b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{58c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{59a}$$
$$4b_{m,n-1}^2 = \left(\sqrt{2r^2 + 2rb_{m,n}}\right)^2, 1 \leq n \leq m \tag{59b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{59c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{60a}$$
$$4b_{m,n-1}^2 = 2r^2 + 2rb_{m,n}, 1 \leq n \leq m \tag{60b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{60c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{61a}$$
$$4b_{m,n-1}^2 - 2r^2 = 2r^2 + 2rb_{m,n} - 2r^2, 1 \leq n \leq m \tag{61b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{61c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{62a}$$
$$4b_{m,n-1}^2 - 2r^2 = 2rb_{m,n}, 1 \leq n \leq m \tag{62b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{62c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{63a}$$
$$\frac{4b_{m,n-1}^2 - 2r^2}{2r} = \frac{2rb_{m,n}}{2r}, 1 \leq n \leq m \tag{63b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{63c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{64a}$$
$$\frac{4b_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2rb_{m,n}}{2r}, 1 \leq n \leq m \tag{64b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{64c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{65a}$$
$$\frac{2b_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2rb_{m,n}}{2r}, 1 \leq n \leq m \tag{65b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{65c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{66a}$$
$$\frac{2b_{m,n-1}^2}{r} - r = \frac{2rb_{m,n}}{2r}, 1 \leq n \leq m \tag{66b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{66c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{67a}$$
$$\frac{2b_{m,n-1}^2}{r} - r = b_{m,n}, 1 \leq n \leq m \tag{67b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{67c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{68a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{68b}$$
$$2^m\sqrt{2r^2 - 2rb_{m,0}} = s, m \geq 0 \tag{68c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{69a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{69b}$$
$$\frac{2^m\sqrt{2r^2 - 2rb_{m,0}}}{2^m} = \frac{s}{2^m}, m \geq 0 \tag{69c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{70a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{70b}$$
$$\sqrt{2r^2 - 2rb_{m,0}} = \frac{s}{2^m}, m \geq 0 \tag{70c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{71a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{71b}$$
$$\left(\sqrt{2r^2 - 2rb_{m,0}}\right)^2 = \left(\frac{s}{2^m}\right)^2, m \geq 0 \tag{71c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{72a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{72b}$$
$$2r^2 - 2rb_{m,0} = \left(\frac{s}{2^m}\right)^2, m \geq 0 \tag{72c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{73a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{73b}$$
$$2r^2 - 2rb_{m,0} = \frac{s^2}{\left(2^m\right)^2}, m \geq 0 \tag{73c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{74a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{74b}$$
$$2r^2 - 2rb_{m,0} = \frac{s^2}{2^{2m}}, m \geq 0 \tag{74c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{75a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{75b}$$
$$2r^2 - 2rb_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0 \tag{75c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{76a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{76b}$$
$$-2rb_{m,0} = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0 \tag{76c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{77a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{77b}$$
$$\frac{-2rb_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}, m \geq 0 \tag{77c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{78a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{78b}$$
$$b_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}, m \geq 0 \tag{78c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{79a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{79b}$$
$$b_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r}, m \geq 0 \tag{79c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{80a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{80b}$$
$$b_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r}, m \geq 0 \tag{80c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{81a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{81b}$$
$$b_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r}, m \geq 0 \tag{81c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{82a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{82b}$$
$$b_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r}, m \geq 0 \tag{82c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{83a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{83b}$$
$$b_{m,0} = \frac{s^2}{-2^{2m+1}r} + r, m \geq 0 \tag{83c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{84a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{84b}$$
$$b_{m,0} = r + \frac{s^2}{-2^{2m+1}r}, m \ge 0 \tag{84c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{85a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{85b}$$
$$b_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r}\right), m \ge 0 \tag{85c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{86a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{86b}$$
$$b_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r}\right), m \ge 0 \tag{86c}$$

$$x = \lim_{m\to\infty} b_{m,m} \tag{87a}$$
$$b_{m,n} = \frac{2b_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{87b}$$
$$b_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{87c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{88a}$$
$$r\hat{b}_{m,n} = \frac{2(r\hat{b}_{m,n-1})^2}{r} - r, 1 \le n \le m \tag{88b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{88c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{89a}$$
$$r\hat{b}_{m,n} = \frac{2r^2\hat{b}_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{89b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{89c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{90a}$$
$$r\hat{b}_{m,n} = 2r\hat{b}_{m,n-1}^2 - r, 1 \le n \le m \tag{90b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{90c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{91a}$$
$$\frac{r\hat{b}_{m,n}}{r} = \frac{2r\hat{b}^2_{m,n-1} - r}{r}, 1 \leq n \leq m \tag{91b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0 \tag{91c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{92a}$$
$$\hat{b}_{m,n} = \frac{2r\hat{b}^2_{m,n-1} - r}{r}, 1 \leq n \leq m \tag{92b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0 \tag{92c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{93a}$$
$$\hat{b}_{m,n} = \frac{2r\hat{b}^2_{m,n-1}}{r} - \frac{r}{r}, 1 \leq n \leq m \tag{93b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0 \tag{93c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{94a}$$
$$\hat{b}_{m,n} = 2\hat{b}^2_{m,n-1} - \frac{r}{r}, 1 \leq n \leq m \tag{94b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0 \tag{94c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{95a}$$
$$\hat{b}_{m,n} = 2\hat{b}^2_{m,n-1} - 1, 1 \leq n \leq m \tag{95b}$$
$$r\hat{b}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0 \tag{95c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{96a}$$
$$\hat{b}_{m,n} = 2\hat{b}^2_{m,n-1} - 1, 1 \leq n \leq m \tag{96b}$$
$$\frac{r\hat{b}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \geq 0 \tag{96c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{97a}$$
$$\hat{b}_{m,n} = 2\hat{b}^2_{m,n-1} - 1, 1 \leq n \leq m \tag{97b}$$
$$\hat{b}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \geq 0 \tag{97c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{98a}$$
$$\hat{b}_{m,n} = 2\hat{b}_{m,n-1}^2 - 1, 1 \leq n \leq m \tag{98b}$$
$$\hat{b}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \geq 0 \tag{98c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{99a}$$
$$\hat{b}_{m,n} = 2\hat{b}_{m,n-1}^2 - 1, 1 \leq n \leq m \tag{99b}$$
$$\hat{b}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \geq 0 \tag{99c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{100a}$$
$$\hat{b}_{m,n} = 2\hat{b}_{m,n-1}^2 - 1, 1 \leq n \leq m \tag{100b}$$
$$\hat{b}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}, m \geq 0 \tag{100c}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{101a}$$
$$\hat{b}_{m,n} = T_2^{\circ n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), 0 \leq n \leq m \tag{101b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{102a}$$
$$\hat{b}_{m,n} = T_{2^n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), 0 \leq n \leq m \tag{102b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{103a}$$
$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(x_0)}{k!}\left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - x_0\right)^k\Bigg|_{x_0=1}, 0 \leq n \leq m \tag{103b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{104a}$$
$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!}\left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - 1\right)^k, 0 \leq n \leq m \tag{104b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{105a}$$
$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}\left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - 1\right)^k, 0 \leq n \leq m \tag{105b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{106a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1\right)^k, 0 \le n \le m \tag{106b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{107a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!} \left(-\frac{s^2}{2^{2m+1}r^2}\right)^k, 0 \le n \le m \tag{107b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{108a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!} \left(\frac{-s^2}{2^{2m+1}r^2}\right)^k, 0 \le n \le m \tag{108b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{109a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \left(\frac{-s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{109b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{110a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(2 \cdot \frac{-s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{110b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{111a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-2s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{111b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{112a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-s^2}{2^{2m}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{112b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{113a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s^2}{r^2}\right)}{2^{2m}}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{113b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{114a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{2^{2m}}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{114b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{115a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{(2^2)^m}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{115b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{116a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{4^m}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{116b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{117a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{\left(-\left(\frac{s}{r}\right)^2\right)^k}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{117b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{118a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k\left(\left(\frac{s}{r}\right)^2\right)^k}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{118b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{119a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \le n \le m \tag{119b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{120a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k}\right)}{(2k)!}, 0 \le n \le m \tag{120b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{121a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k (2k)!}, 0 \le n \le m \tag{121b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{122a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!(4^m)^k}, 0 \le n \le m \tag{122b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{123a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k}, 0 \le n \le m \tag{123b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{124a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1}((2^n)^2 - j^2)}{\prod_{j=0}^{k-1} 4^m}, 0 \le n \le m \tag{124b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{125a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, 0 \le n \le m \tag{125b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{126a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{4^m}, 0 \le n \le m \tag{126b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{127a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^2)^n - j^2}{4^m}, 0 \le n \le m \tag{127b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{128a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m}, 0 \le n \le m \tag{128b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{129a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{129b}$$

$$x = \lim_{m\to\infty} r\hat{b}_{m,m} \tag{130a}$$

$$\hat{b}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{130b}$$

$$x = \lim_{m\to\infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m}\right) \tag{131}$$

$$x = \lim_{m\to\infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \tag{132}$$

$$x = \lim_{m\to\infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \tag{133}$$

$$0 \leq j \leq k-1 < 2^m \tag{134}$$

$$0 \leq j < 2^m \tag{135}$$

$$0^2 \leq j^2 < (2^m)^2 \tag{136}$$

$$0 \leq j^2 < (2^m)^2 \tag{137}$$

$$0 \leq j^2 < 2^{2m} \tag{138}$$

$$0 \leq j^2 < (2^2)^m \tag{139}$$

$$0 \leq j^2 < 4^m \tag{140}$$

$$\frac{0}{4^m} \leq \frac{j^2}{4^m} < \frac{4^m}{4^m} \tag{141}$$

$$0 \leq \frac{j^2}{4^m} < \frac{4^m}{4^m} \tag{142}$$

$$0 \leq \frac{j^2}{4^m} < 1 \tag{143}$$

$$-0 \geq -\frac{j^2}{4^m} > -1 \tag{144}$$

$$0 \geq -\frac{j^2}{4^m} > -1 \tag{145}$$

$$0 + 1 \geq -\frac{j^2}{4^m} + 1 > -1 + 1 \tag{146}$$

$$1 \geq -\frac{j^2}{4^m} + 1 > -1 + 1 \tag{147}$$

$$1 \geq 1 - \frac{j^2}{4^m} > -1 + 1 \tag{148}$$

$$1 \geq 1 - \frac{j^2}{4^m} > 0 \tag{149}$$

$$0 < 1 - \frac{j^2}{4^m} \leq 1 \tag{150}$$

$$0 < \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \leq 1 \tag{151}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|(-1)^k \left(\frac{s}{r}\right)^{2k}\right|}{(2k)!} \tag{152}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|(-1)^k\right| \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{153}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{154}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{\left(\frac{-\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| \tag{155}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{\left(\frac{-\left(\frac{s}{r}\right)^{2k+2}}{(2(k+1))!}\right)}{\left(\frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| \tag{156}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{\left(\frac{-\left(\frac{s}{r}\right)^{2k+2}}{(2k+2)!}\right)}{\left(\frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| \tag{157}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{\left(\frac{-\left(\frac{s}{r}\right)^{2k+2}}{(2k+2)(2k+1)(2k)!}\right)}{\left(\frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| \tag{158}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{\left(\frac{-\left(\frac{s}{r}\right)^{2k+2}}{(2k+2)(2k+1)}\right)}{\left(\frac{s}{r}\right)^{2k}}\right| \tag{159}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{-\left(\frac{s}{r}\right)^{2k+2-2k}}{(2k+2)(2k+1)}\right| \tag{160}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \left|\frac{-\left(\frac{s}{r}\right)^{2}}{(2k+2)(2k+1)}\right| \tag{161}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \frac{\left|-\left(\frac{s}{r}\right)^{2}\right|}{(2k+2)(2k+1)} \tag{162}$$

$$\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \frac{\left(\frac{s}{r}\right)^{2}}{(2k+2)(2k+1)} \tag{163}$$

$$\lim_{k\to\infty}\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = \lim_{k\to\infty}\frac{\left(\frac{s}{r}\right)^{2}}{(2k+2)(2k+1)} \tag{164}$$

$$\lim_{k\to\infty}\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| = 0 \tag{165}$$

$$\lim_{k\to\infty}\left|\frac{\left(\frac{(-1)^{k+1}\left(\frac{s}{r}\right)^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}\right| < 1 \tag{166}$$

$$\sum_{k=0}^{\infty}\frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!} < \infty \tag{167}$$

$$\left|\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^{2}}{4^{m}}\right)\right| \leq \frac{\left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{168}$$

$$x = r\sum_{k=0}^{\infty}\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\lim_{m\to\infty}\prod_{j=0}^{k-1}\left(1-\frac{j^{2}}{4^{m}}\right) \tag{169}$$

$$x = r\sum_{k=0}^{\infty}\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}(1-0) \tag{170}$$

$$x = r\sum_{k=0}^{\infty}\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}1 \tag{171}$$

$$x = r\sum_{k=0}^{\infty}\frac{(-1)^{k}\left(\frac{s}{r}\right)^{2k}}{(2k)!}\cdot 1^{k} \tag{172}$$

$$x = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \cdot 1 \tag{173}$$

$$x = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{174}$$

$$\cos(r, s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{175}$$

0.1.4.1.1 Negation

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{-s}{r}\right)^{2k}}{(2k)!} \tag{176}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{s}{r}\right)^{2k}}{(2k)!} \tag{177}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k (-1)^{2k} \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{178}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^{k+2k} \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{179}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^{3k} \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{180}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{((-1)^3)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{181}$$

$$\cos(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{182}$$

$$\cos(r, -s) = \cos(r, s) \tag{183}$$

0.1.4.1.2 Subtraction

$$\sin^2(r,\alpha-\beta)+(r-\cos(r,\alpha-\beta))^2=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{184}$$

$$\cos^2(r,\alpha-\beta)+\sin^2(r,\alpha-\beta)+(r-\cos(r,\alpha-\beta))^2-\cos^2(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{185}$$

$$r^2+(r-\cos(r,\alpha-\beta))^2-\cos^2(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{186}$$

$$r^2+r^2-2r\cos(r,\alpha-\beta)+\cos^2(r,\alpha-\beta)-\cos^2(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{187}$$

$$2r^2-2r\cos(r,\alpha-\beta)+\cos^2(r,\alpha-\beta)-\cos^2(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{188}$$

$$2r^2-2r\cos(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2 \tag{189}$$

$$2r^2-2r\cos(r,\alpha-\beta)-2r^2=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2-2r^2 \tag{190}$$

$$-2r\cos(r,\alpha-\beta)=(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2-2r^2 \tag{191}$$

$$\frac{-2r\cos(r,\alpha-\beta)}{-2r}=\frac{(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2-2r^2}{-2r} \tag{192}$$

$$\cos(r,\alpha-\beta)=\frac{(\cos(r,\alpha)-\cos(r,\beta))^2+(\sin(r,\alpha)-\sin(r,\beta))^2-2r^2}{-2r} \tag{193}$$

$$\cos(r,\alpha-\beta)=\frac{\cos^2(r,\alpha)-2\cos(r,\alpha)\cos(r,\beta)+\cos^2(r,\beta)+(\sin(r,\alpha)-\sin(r,\beta))^2-2r^2}{-2r} \tag{194}$$

$$\cos(r,\alpha-\beta)=\frac{\cos^2(r,\alpha)-2\cos(r,\alpha)\cos(r,\beta)+\cos^2(r,\beta)+\sin^2(r,\alpha)-2\sin(r,\alpha)\sin(r,\beta)+\sin^2(r,\beta)-2r^2}{-2r} \tag{195}$$

$$\cos(r,\alpha-\beta)=\frac{(\cos^2(r,\alpha)+\sin^2(r,\alpha))-2\cos(r,\alpha)\cos(r,\beta)+\cos^2(r,\beta)-2\sin(r,\alpha)\sin(r,\beta)+\sin^2(r,\beta)-2r^2}{-2r} \tag{196}$$

$$\cos(r,\alpha-\beta)=\frac{(\cos^2(r,\alpha)+\sin^2(r,\alpha))-2\cos(r,\alpha)\cos(r,\beta)+(\cos^2(r,\beta)+\sin^2(r,\beta))-2\sin(r,\alpha)\sin(r,\beta)-2r^2}{-2r} \tag{197}$$

$$\cos(r,\alpha-\beta)=\frac{r^2-2\cos(r,\alpha)\cos(r,\beta)+(\cos^2(r,\beta)+\sin^2(r,\beta))-2\sin(r,\alpha)\sin(r,\beta)-2r^2}{-2r} \tag{198}$$

$$\cos(r,\alpha-\beta)=\frac{r^2-2\cos(r,\alpha)\cos(r,\beta)+r^2-2\sin(r,\alpha)\sin(r,\beta)-2r^2}{-2r} \tag{199}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - 2\cos(r, \alpha)\cos(r, \beta) - 2\sin(r, \alpha)\sin(r, \beta) - 2r^2}{-2r} \tag{200}$$

$$\cos(r, \alpha - \beta) = \frac{-2\cos(r, \alpha)\cos(r, \beta) - 2\sin(r, \alpha)\sin(r, \beta)}{-2r} \tag{201}$$

$$\cos(r, \alpha - \beta) = \frac{-2\left(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta)\right)}{-2r} \tag{202}$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta)}{r} \tag{203}$$

0.1.4.1.3 Addition

$$\cos(r, \alpha + \beta) = \cos(r, \alpha - (-\beta)) \tag{204}$$

$$\cos(r, \alpha + \beta) = \frac{\cos(r, \alpha)\cos(r, -\beta) + \sin(r, \alpha)\sin(r, -\beta)}{r} \tag{205}$$

$$\cos(r, \alpha + \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, -\beta)}{r} \tag{206}$$

$$\cos(r, \alpha + \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)(-\sin(r, \beta))}{r} \tag{207}$$

$$\cos(r, \alpha + \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) - \sin(r, \alpha)\sin(r, \beta)}{r} \tag{208}$$

0.1.4.1.4 Periodicity

$$\cos(r, s + Q_r) = \frac{\cos(r, s)\cos(r, Q_r) - \sin(r, s)\sin(r, Q_r)}{r} \tag{209}$$

$$\cos(r, s + Q_r) = \frac{\cos(r, s) \cdot 0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{210}$$

$$\cos(r, s + Q_r) = \frac{0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{211}$$

$$\cos(r, s + Q_r) = \frac{-\sin(r, s)\sin(r, Q_r)}{r} \tag{212}$$

$$\cos(r, s + Q_r) = \frac{-\sin(r, s) \cdot r}{r} \tag{213}$$

$$\cos(r, s + Q_r) = -\sin(r, s) \tag{214}$$

$$\cos(r, s + Q_r + Q_r) = -\sin(r, s + Q_r) \tag{215}$$

$$\cos(r, s + 2Q_r) = -\sin(r, s + Q_r) \tag{216}$$

$$\cos(r, s + 2Q_r) = -\frac{\sin(r, s)\cos(r, Q_r) + \cos(r, s)\sin(r, Q_r)}{r} \tag{217}$$

$$\cos(r, s + 2Q_r) = -\frac{\sin(r, s) \cdot 0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{218}$$

$$\cos(r, s + 2Q_r) = -\frac{0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{219}$$

$$\cos(r, s + 2Q_r) = -\frac{\cos(r, s)\sin(r, Q_r)}{r} \tag{220}$$

$$\cos(r, s + 2Q_r) = -\frac{\cos(r, s) \cdot r}{r} \tag{221}$$

$$\cos(r, s + 2Q_r) = \frac{-\cos(r, s) \cdot r}{r} \tag{222}$$

$$\cos(r, s + 2Q_r) = -\cos(r, s) \tag{223}$$

$$\cos(r, s + 2Q_r + Q_r) = -\cos(r, s + Q_r) \tag{224}$$

$$\cos(r, s + 3Q_r) = -\cos(r, s + Q_r) \tag{225}$$

$$\cos(r, s + 3Q_r) = -(-\sin(r, s)) \tag{226}$$

$$\cos(r, s + 3Q_r) = \sin(r, s) \tag{227}$$

$$\cos(r, s + 3Q_r + Q_r) = \sin(r, s + Q_r) \tag{228}$$

$$\cos(r, s + 4Q_r) = \sin(r, s + Q_r) \tag{229}$$

$$\cos(r, s + 4Q_r) = \frac{\sin(r, s)\cos(r, Q_r) + \cos(r, s)\sin(r, Q_r)}{r} \tag{230}$$

$$\cos(r, s + 4Q_r) = \frac{\sin(r, s) \cdot 0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{231}$$

$$\cos(r, s + 4Q_r) = \frac{0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{232}$$

$$\cos(r, s + 4Q_r) = \frac{\cos(r, s)\sin(r, Q_r)}{r} \tag{233}$$

$$\cos(r, s + 4Q_r) = \frac{\cos(r, s) \cdot r}{r} \tag{234}$$

$$\cos(r, s + 4Q_r) = \cos(r, s) \tag{235}$$

0.1.4.1.5 Partial Derivative

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \tag{236}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \tag{237}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k}(2k)!} \right)}{\partial s} \tag{238}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k}(2k)!} \right)}{\partial s} \tag{239}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1}(2k)!} \right)}{\partial s} \tag{240}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1}(2k)!} \tag{241}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1}(2k)!} \tag{242}$$

$$\frac{\partial \cos(r, s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \tag{243}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \tag{244}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \tag{245}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{246}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{247}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{248}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \tag{249}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \tag{250}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=0}^{\infty} \frac{(-1)^{k} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{251}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{252}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-\sin(r,s)}{r} \tag{253}$$

0.1.4.2 Sine

$$\begin{aligned} \sin\colon \mathbb{R}^+ \times \mathbb{R} &\to \mathbb{R} \\ (r, s) &\mapsto y \in [-r, r] \end{aligned} \tag{254}$$

$$(\alpha, \beta) \in \mathbb{R}^2 \tag{255}$$

$$\cos(r, \alpha - \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha-\beta}{r}\right)^{2k}}{(2k)!} \tag{256}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha-\beta}{r}\right)^{2k}}{(2k)!} \tag{257}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r} - \frac{\beta}{r}\right)^{2k}}{(2k)!} \tag{258}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \left(\frac{\alpha}{r} + \frac{-\beta}{r}\right)^{2k}}{(2k)!} \tag{259}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^{j}}{(2k)!} \tag{260}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^{j}}{(2k)!} \tag{261}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{262}$$

$$\cos(r, \alpha - \beta) = r \lim_{n\to\infty} \sum_{k=0}^{n} \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{263}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!} \tag{264}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{(2k-j)!j!} \tag{265}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}}{(2k-j)!}\frac{\left(\frac{-\beta}{r}\right)^{j}}{j!} \tag{266}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{j=0}^{n-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j}}{(2j)!}+\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right) \tag{267}$$

$$\cos(r,\alpha-\beta)=r\left(\lim_{n\to\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{j=0}^{n-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j}}{(2j)!}\right)+\lim_{n\to\infty}\left(\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right)\right) \tag{268}$$

$$\cos(r,\alpha-\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+\lim_{n\to\infty}\left(\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right)\right) \tag{269}$$

$$\cos(r,\alpha-\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{270}$$

$$\cos(r,\alpha-\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{271}$$

$$\cos(r,\alpha-\beta)=\frac{r\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{272}$$

$$\cos(r,\alpha-\beta)=\frac{r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{273}$$

$$\cos(r,\alpha-\beta)=\frac{\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{274}$$

$$\cos(r,\alpha-\beta)=\frac{\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{275}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{276}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{277}$$

$$\frac{\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)}{r}=\frac{\cos(r,\alpha)\cos(r,\beta)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{278}$$

$$\sin(r,\alpha)\sin(r,\beta)=\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{279}$$

$$\sin(r,s)\overset{\{\alpha,\beta\}\mapsto s}{=}r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{280}$$

0.1.4.2.1 Negation

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{-s}{r}\right)^{2k+1}}{(2k+1)!} \tag{281}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(-\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{282}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^k (-1)^{2k+1} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{283}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^{k+2k+1} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{284}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{(-1)^{3k+1} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{285}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{-(-1)^{3k} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{286}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{-((-1)^3)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{287}$$

$$\sin(r, -s) = r \sum_{k=0}^{\infty} \frac{-(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{288}$$

$$\sin(r, -s) = -r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{289}$$

$$\sin(r, -s) = -\sin(r, s) \tag{290}$$

0.1.4.2.2 Addition/Subtraction

$$\sin(r, \alpha \pm \beta) = \cos(r, Q_r - (\alpha \pm \beta)) \tag{291}$$

$$\sin(r, \alpha \pm \beta) = \cos(r, Q_r - \alpha \mp \beta) \tag{292}$$

$$\sin(r, \alpha \pm \beta) = \cos(r, (Q_r - \alpha) \mp \beta) \tag{293}$$

$$\sin(r, \alpha \pm \beta) = \frac{\cos(r, Q_r - \alpha)\cos(r, \beta) \pm \sin(r, Q_r - \alpha)\sin(r, \beta)}{r} \tag{294}$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha)\cos(r, \beta) \pm \sin(r, Q_r - \alpha)\sin(r, \beta)}{r} \tag{295}$$

$$\sin(r, \alpha \pm \beta) = \frac{\sin(r, \alpha)\cos(r, \beta) \pm \cos(r, \alpha)\sin(r, \beta)}{r} \tag{296}$$

0.1.4.2.3 Periodicity

$$\sin(r, s + Q_r) = \frac{\sin(r, s)\cos(r, Q_r) + \cos(r, s)\sin(r, Q_r)}{r} \tag{297}$$

$$\sin(r, s + Q_r) = \frac{\sin(r, s) \cdot 0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{298}$$

$$\sin(r, s + Q_r) = \frac{0 + \cos(r, s)\sin(r, Q_r)}{r} \tag{299}$$

$$\sin(r, s + Q_r) = \frac{\cos(r, s)\sin(r, Q_r)}{r} \tag{300}$$

$$\sin(r, s + Q_r) = \frac{\cos(r, s) \cdot r}{r} \tag{301}$$

$$\sin(r, s + Q_r) = \cos(r, s) \tag{302}$$

$$\sin(r, s + Q_r + Q_r) = \cos(r, s + Q_r) \tag{303}$$

$$\sin(r, s + 2Q_r) = \cos(r, s + Q_r) \tag{304}$$

$$\sin(r, s + 2Q_r) = \frac{\cos(r, s)\cos(r, Q_r) - \sin(r, s)\sin(r, Q_r)}{r} \tag{305}$$

$$\sin(r, s + 2Q_r) = \frac{\cos(r, s) \cdot 0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{306}$$

$$\sin(r, s + 2Q_r) = \frac{0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{307}$$

$$\sin(r, s + 2Q_r) = \frac{-\sin(r, s)\sin(r, Q_r)}{r} \tag{308}$$

$$\sin(r, s + 2Q_r) = \frac{-\sin(r, s) \cdot r}{r} \tag{309}$$

$$\sin(r, s + 2Q_r) = -\sin(r, s) \tag{310}$$

$$\sin(r, s + 2Q_r + Q_r) = -\sin(r, s + Q_r) \tag{311}$$

$$\sin(r, s + 3Q_r) = -\sin(r, s + Q_r) \tag{312}$$

$$\sin(r, s + 3Q_r) = -\cos(r, s) \tag{313}$$

$$\sin(r, s + 3Q_r + Q_r) = -\cos(r, s + Q_r) \tag{314}$$

$$\sin(r, s + 4Q_r) = -\cos(r, s + Q_r) \tag{315}$$

$$\sin(r, s + 4Q_r) = -\frac{\cos(r, s)\cos(r, Q_r) - \sin(r, s)\sin(r, Q_r)}{r} \tag{316}$$

$$\sin(r, s + 4Q_r) = -\frac{\cos(r, s) \cdot 0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{317}$$

$$\sin(r, s + 4Q_r) = -\frac{0 - \sin(r, s)\sin(r, Q_r)}{r} \tag{318}$$

$$\sin(r, s + 4Q_r) = -\frac{-\sin(r, s)\sin(r, Q_r)}{r} \tag{319}$$

$$\sin(r, s + 4Q_r) = -\frac{-\sin(r, s) \cdot r}{r} \tag{320}$$

$$\sin(r, s + 4Q_r) = \frac{-(-\sin(r, s) \cdot r)}{r} \tag{321}$$

$$\sin(r, s + 4Q_r) = \frac{-(-\sin(r, s)) \cdot r}{r} \tag{322}$$

$$\sin(r, s + 4Q_r) = \frac{\sin(r, s) \cdot r}{r} \tag{323}$$

$$\sin(r, s + 4Q_r) = \sin(r, s) \tag{324}$$

0.1.4.2.4 Partial Derivative

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \tag{325}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \tag{326}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{327}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{328}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k}(2k+1)!} \right)}{\partial s} \tag{329}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k}(2k+1)!} \tag{330}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k}(2k+1)!} \tag{331}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \tag{332}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \tag{333}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \tag{334}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{335}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \tag{336}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\cos(r, s)}{r} \tag{337}$$

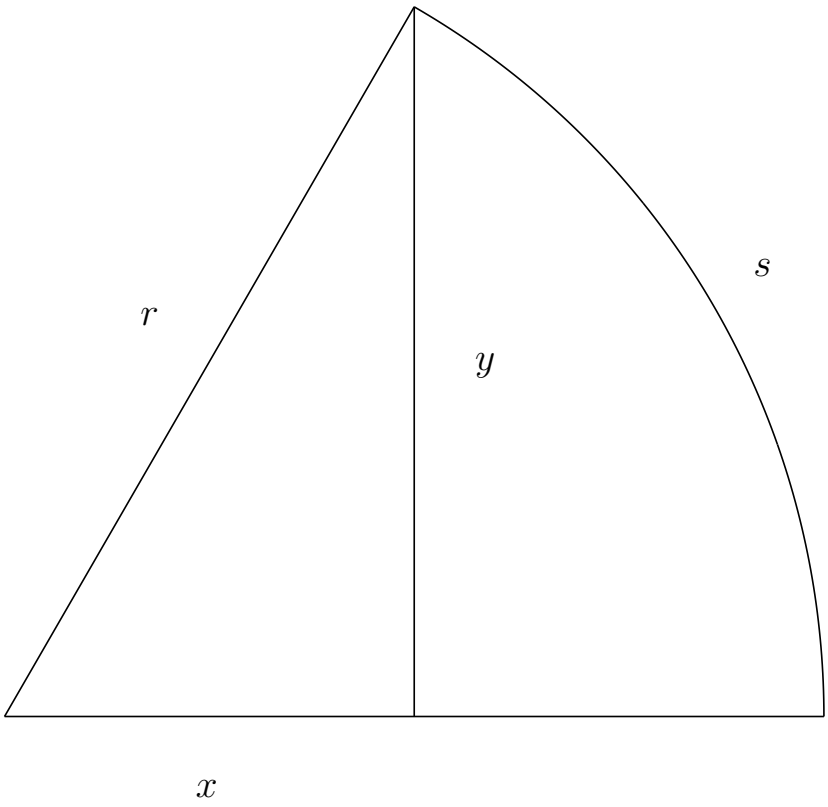


Figure 1: Circular sector.

0.2 Circular Sector Sequence

0.2.1 Apothem

$$\text{area}\triangle OAC_{n-1}=\frac{\overline{AC_{n-1}OB_n}}{2},n\geq 1 \tag{338}$$

$$\text{area}\triangle OAC_{n-1}=\frac{\overline{OC_{n-1}AB_{n-1}}}{2},n\geq 1 \tag{339}$$

$$\frac{\overline{AC_{n-1}OB_n}}{2}=\frac{\overline{OC_{n-1}AB_{n-1}}}{2},n\geq 1 \tag{340}$$

$$\overline{AC_{n-1}OB_n}=\overline{OC_{n-1}AB_{n-1}},n\geq 1 \tag{341}$$

$$\frac{\overline{AC_{n-1}OB_n}}{\overline{AC_{n-1}}}=\frac{\overline{OC_{n-1}AB_{n-1}}}{\overline{AC_{n-1}}},n\geq 1 \tag{342}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}AB_{n-1}}}{\overline{AC_{n-1}}},n\geq 1 \tag{343}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}}\sqrt{\overline{OA}^2-\overline{OB_{n-1}}^2}}{\overline{AC_{n-1}}},n\geq 1 \tag{344}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}}\sqrt{\overline{OA}^2-\overline{OB_{n-1}}^2}}{\sqrt{\overline{OA}^2+\overline{OC_{n-1}}^2-2\overline{OC_{n-1}OB_{n-1}}}},n\geq 1 \tag{345}$$

0.2.2 Half-chord Length

$$\overline{OB_n}^2+\overline{AB_n}^2=\overline{OA}^2 \tag{346}$$

$$\overline{OB_n}^2+\overline{AB_n}^2-\overline{OB_n}^2=\overline{OA}^2-\overline{OB_n}^2 \tag{347}$$

$$\overline{AB_n}^2=\overline{OA}^2-\overline{OB_n}^2 \tag{348}$$

$$\overline{AB_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2} \tag{349}$$

0.2.3 Chord Length

$$\overline{AC_n}^2=\overline{AB_n}^2+\overline{B_nC_n}^2 \tag{350}$$

$$\overline{AC_n}=\sqrt{\overline{AB_n}^2+\overline{B_nC_n}^2} \tag{351}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\overline{B_nC_n}^2} \tag{352}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\left(\overline{OC_n}-\overline{OB_n}\right)^2} \tag{353}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\overline{OC_n}^2-2\overline{OC_nOB_n}+\overline{OB_n}^2} \tag{354}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2+\overline{OC_n}^2-2\overline{OC_nOB_n}} \tag{355}$$

0.2.4 Arc Length

$$\operatorname{arc} AC_0 \stackrel{\text{def}}{=} \lim_{n\to\infty} 2^n \overline{AC_n} \tag{356}$$

0.2.5 Area

$$\operatorname{area} \triangledown OAC_0 = \lim_{n\to\infty} 2^{n-1} \operatorname{area} OAC_nC_{n-1} \tag{357}$$

$$\operatorname{area} \triangledown OAC_0 = \lim_{n\to\infty} 2^{n-1} \frac{\overline{AC_{n-1}OC_n}}{2} \tag{358}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} 2^{n-1} \overline{AC_{n-1}OC_n}}{2} \tag{359}$$

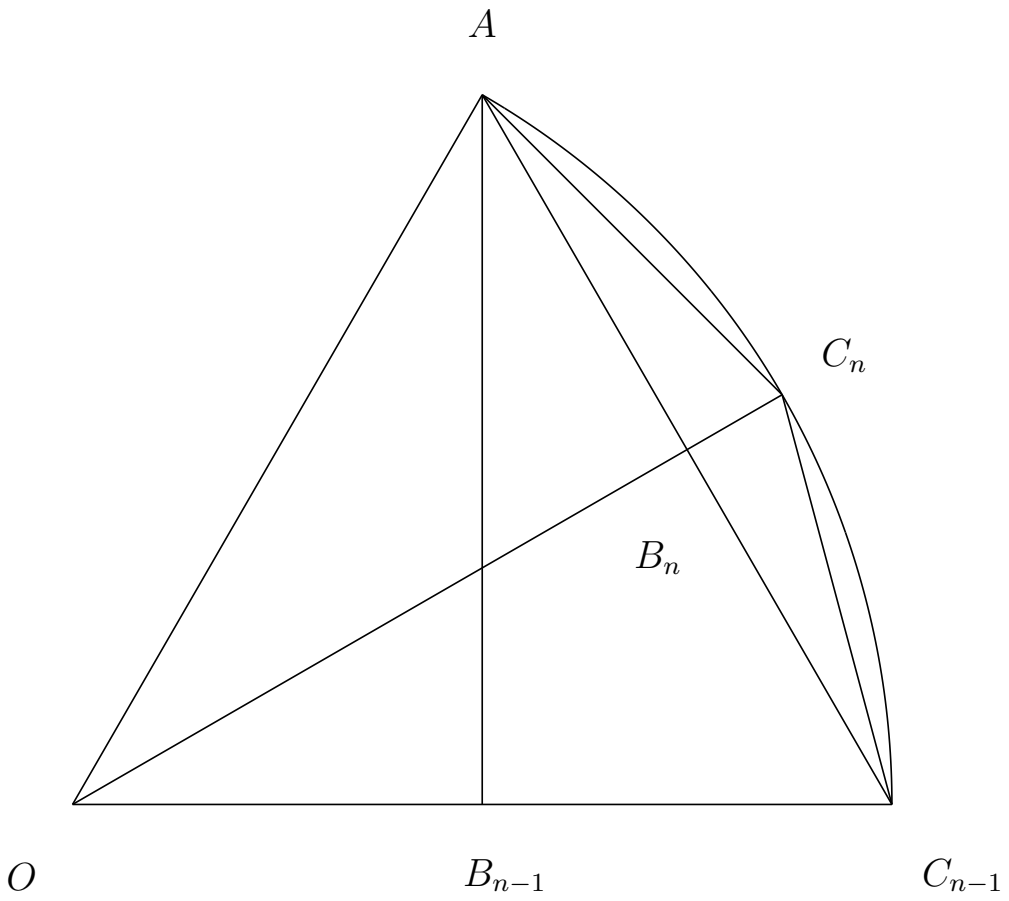
$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} \left(2^{n-1} \overline{AC_{n-1}}\right) \overline{OC_n}}{2} \tag{360}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} \overline{OC_n} \left(2^{n-1} \overline{AC_{n-1}}\right)}{2} \tag{361}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\left(\lim_{n\to\infty} \overline{OC_n}\right) \left(\lim_{n\to\infty} 2^{n-1} \overline{AC_{n-1}}\right)}{2} \tag{362}$$

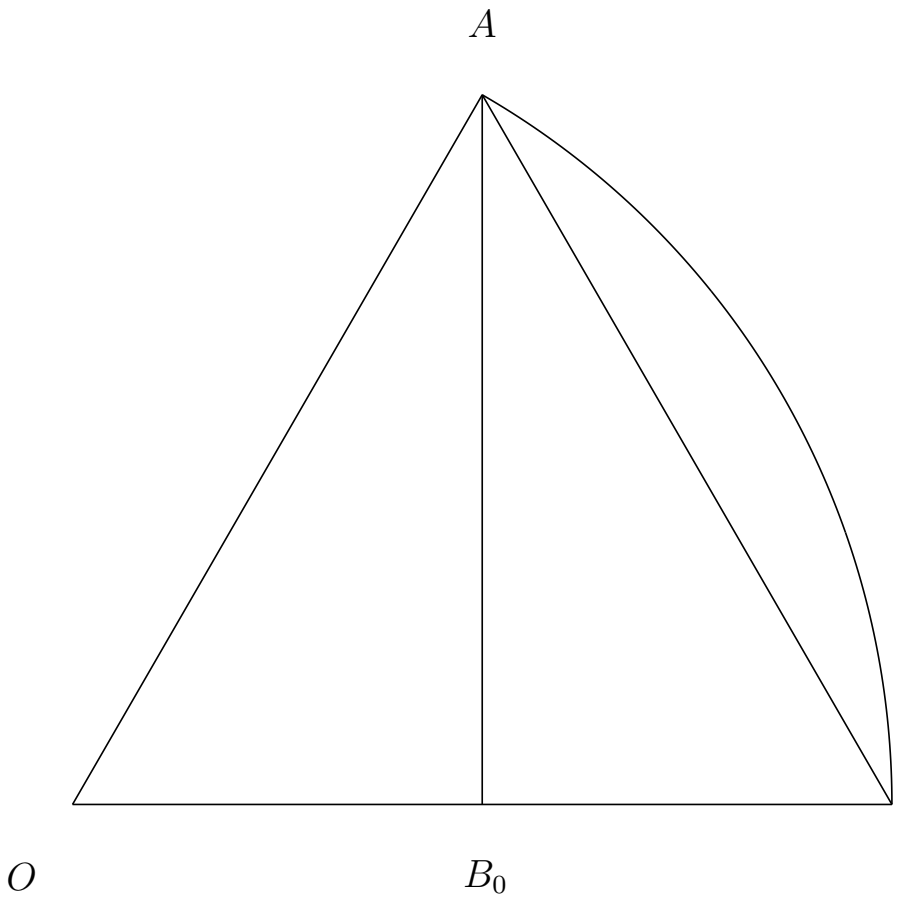
$$\operatorname{area} \triangledown OAC_0 = \frac{\overline{OA} \lim_{n\to\infty} 2^{n-1} \overline{AC_{n-1}}}{2} \tag{363}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\overline{OA} \operatorname{arc} AC_0}{2} \tag{364}$$



(a) Recursive step.

Figure 2: Circular sector sequence.



(b) Base case.

Figure 2: Circular sector sequence.

0.3 Simple Harmonic Motion

$$[m] = \text{kg} \tag{365}$$

$$[\omega] = \text{m/s} \tag{366}$$

$$[k] = \text{N/m} \tag{367}$$

$$[r] = \text{m} \tag{368}$$

$$T = 4Q_r \tag{369}$$

$$s = \omega t \tag{370}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta s|}{|\Delta t|} \tag{371}$$

$$(\Delta s)^2 \approx (\Delta x)^2 + (\Delta y)^2 \tag{372}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx \frac{(\Delta x)^2}{(\Delta x)^2} + \frac{(\Delta y)^2}{(\Delta x)^2} \tag{373}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx 1 + \frac{(\Delta y)^2}{(\Delta x)^2} \tag{374}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx 1 + \left(\frac{\Delta y}{\Delta x}\right)^2 \tag{375}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} = 1 + 0 \iff x \to 0 \tag{376}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} = 1 \iff x \to 0 \tag{377}$$

$$(\Delta x)^2 \frac{(\Delta s)^2}{(\Delta x)^2} = (\Delta x)^2 \iff x \to 0 \tag{378}$$

$$\frac{(\Delta x)^2 (\Delta s)^2}{(\Delta x)^2} = (\Delta x)^2 \iff x \to 0 \tag{379}$$

$$(\Delta s)^2 = (\Delta x)^2 \iff x \to 0 \tag{380}$$

$$\sqrt{(\Delta s)^2} = \sqrt{(\Delta x)^2} \iff x \to 0 \tag{381}$$

$$|\Delta s| = \sqrt{(\Delta x)^2} \iff x \to 0 \tag{382}$$

$$|\Delta s| = |\Delta x| \iff x \to 0 \tag{383}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta x|}{|\Delta t|} \iff x \to 0 \tag{384}$$

$$F = ma \tag{385}$$

$$F = -kx \tag{386}$$

$$ma = -kx \tag{387}$$

$$x = \cos(r, s) \tag{388}$$

$$x(t) = \cos(r, \omega t) \tag{389}$$

$$v(t) = \frac{dx}{dt} \tag{390}$$

$$v(t) = \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{391}$$

$$v(t) = \frac{-\sin(r,\omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{392}$$

$$v(t) = \frac{-\sin(r,\omega t)}{r} \cdot \omega \tag{393}$$

$$v(t) = \frac{-\omega \sin(r,\omega t)}{r} \tag{394}$$

$$a(t) = \frac{dv}{dt} \tag{395}$$

$$a(t) = \frac{d\left(\left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \frac{d(\omega t)}{dt}\right)}{dt} \tag{396}$$

$$a(t) = \left(\frac{\partial^2 \cos(r,s)}{\partial s^2} \cdot \left.\frac{ds}{dt}\right|_{s=\omega t}\right) \cdot \frac{d(\omega t)}{dt} + \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{397}$$

$$a(t) = \left(\frac{-\cos(r,\omega t)}{r^2} \cdot \frac{d(\omega t)}{dt}\right) \cdot \frac{d(\omega t)}{dt} + \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{398}$$

$$a(t) = \frac{-\cos(r,\omega t)}{r^2} \cdot \left(\frac{d(\omega t)}{dt}\right)^2 + \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{399}$$

$$a(t) = \frac{-\cos(r,\omega t)}{r^2} \cdot \omega^2 + \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{400}$$

$$a(t) = \frac{-\omega^2 \cos(r,\omega t)}{r^2} + \left.\frac{\partial \cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{401}$$

$$a(t) = \frac{-\omega^2 \cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r} \cdot \frac{d^2(\omega t)}{dt^2} \tag{402}$$

$$a(t) = \frac{-\omega^2 \cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r} \cdot 0 \tag{403}$$

$$a(t) = \frac{-\omega^2 \cos(r,\omega t)}{r^2} + 0 \tag{404}$$

$$a(t) = \frac{-\omega^2 \cos(r,\omega t)}{r^2} \tag{405}$$

$$a(x) = \frac{-\omega^2 x}{r^2} \tag{406}$$

$$m\frac{-\omega^2 x}{r^2} = -kx \tag{407}$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx \tag{408}$$

$$\frac{-m\omega^2 x}{r^2} = -kx \tag{409}$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2} \tag{410}$$

$$m\omega^2 = kr^2 \tag{411}$$

$$K(t) = \frac{mv^2(t)}{2} \tag{412}$$

$$K(t) = \frac{m\left(\frac{-\omega \sin(r,\omega t)}{r}\right)^2}{2} \tag{413}$$

$$K(t) = \frac{m\frac{(-\omega \sin(r,\omega t))^2}{r^2}}{2} \tag{414}$$

$$K(t) = \frac{m\frac{(-\omega)^2 \sin^2(r,\omega t)}{r^2}}{2} \tag{415}$$

$$K(t) = \frac{m\frac{\omega^2 \sin^2(r,\omega t)}{r^2}}{2} \tag{416}$$

$$K(t) = \frac{\left(\frac{m\omega^2 \sin^2(r,\omega t)}{r^2}\right)}{2} \tag{417}$$

$$K(t) = \frac{\left(\frac{kr^2 \sin^2(r,\omega t)}{r^2}\right)}{2} \tag{418}$$

$$K(t) = \frac{k \sin^2(r,\omega t)}{2} \tag{419}$$

$$U(t) = \frac{kx^2(t)}{2} \tag{420}$$

$$U(t) = \frac{k \cos^2(r,\omega t)}{2} \tag{421}$$

$$E = K + U \tag{422}$$

$$E = \frac{k \sin^2(r,\omega t)}{2} + \frac{k \cos^2(r,\omega t)}{2} \tag{423}$$

$$E = \frac{k \sin^2(r,\omega t) + k \cos^2(r,\omega t)}{2} \tag{424}$$

$$E = \frac{k(\sin^2(r,\omega t) + \cos^2(r,\omega t))}{2} \tag{425}$$

$$E = \frac{kr^2}{2} \tag{426}$$

$$E = \frac{m\omega^2}{2} \tag{427}$$

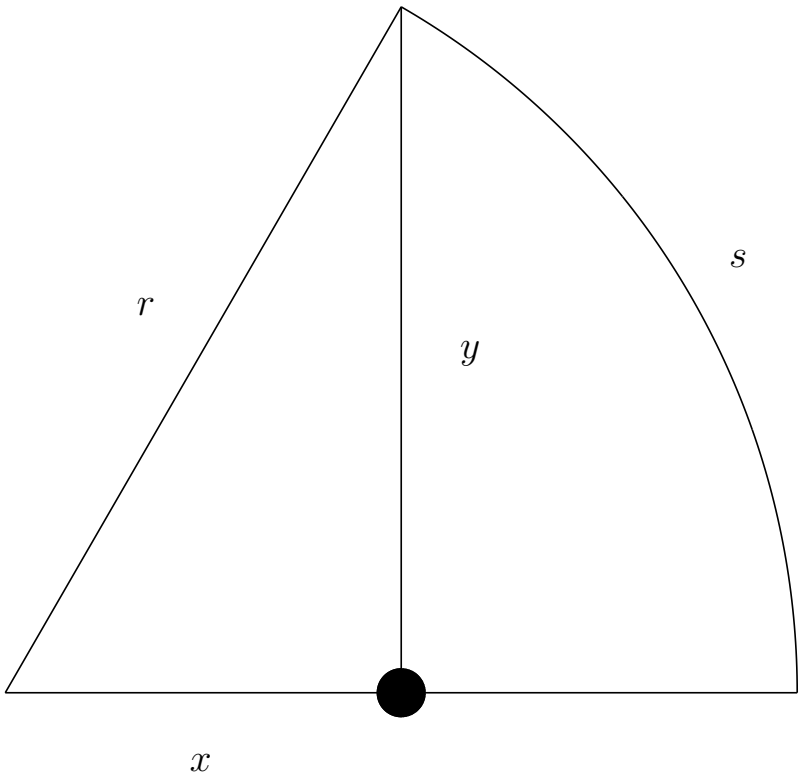


Figure 3: Simple harmonic motion.

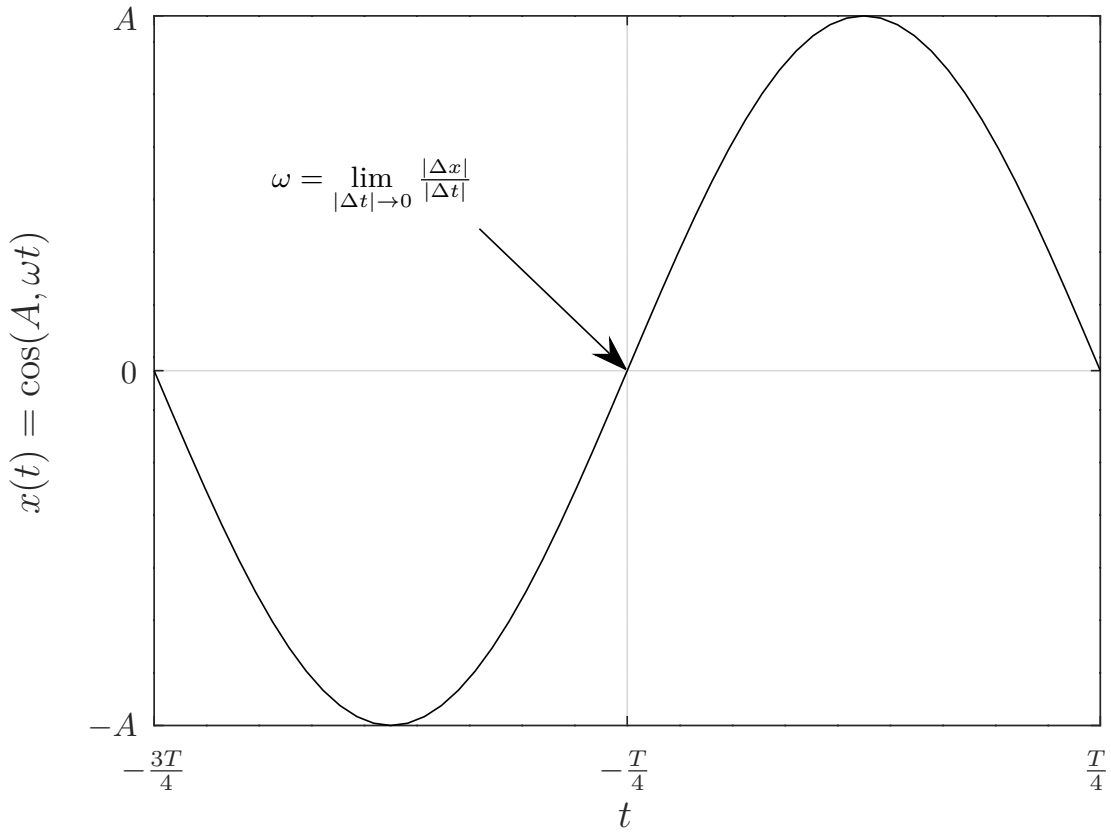


Figure 4: Cosine wave.

0.4 Uniform Circular Motion

$$[m] = \text{kg} \tag{428}$$

$$[\omega] = \text{m/s} \tag{429}$$

$$[r] = \text{m} \tag{430}$$

$$T = 4Q_r \tag{431}$$

$$s = \omega t \tag{432}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta s|}{|\Delta t|} \tag{433}$$

$$x = \cos(r, s) \tag{434}$$

$$x(t) = \cos(r, \omega t) \tag{435}$$

$$y = \sin(r, s) \tag{436}$$

$$y(t) = \sin(r, \omega t) \tag{437}$$

$$\vec{r}(t) = (x, y) \tag{438}$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t)) \tag{439}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \tag{440}$$

$$\vec{v}(t) = \frac{d(x, y)}{dt} \tag{441}$$

$$\vec{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt} \right) \tag{442}$$

$$\frac{dx}{dt} = \left. \frac{\partial \cos(r, s)}{\partial s} \right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{443}$$

$$\frac{dx}{dt} = \frac{-\sin(r, \omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{444}$$

$$\frac{dx}{dt} = \frac{-\sin(r, \omega t)}{r} \cdot \omega \tag{445}$$

$$\frac{dx}{dt} = \frac{-\omega \sin(r, \omega t)}{r} \tag{446}$$

$$\frac{dy}{dt} = \left. \frac{\partial \sin(r, s)}{\partial s} \right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{447}$$

$$\frac{dy}{dt} = \frac{\cos(r, \omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{448}$$

$$\frac{dy}{dt} = \frac{\cos(r, \omega t)}{r} \cdot \omega \tag{449}$$

$$\frac{dy}{dt} = \frac{\omega \cos(r, \omega t)}{r} \tag{450}$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right) \tag{451}$$

$$\vec{v}(t) = \frac{\omega\left(-\sin(r,\omega t),\cos(r,\omega t)\right)}{r} \tag{452}$$

$$v = |\vec{v}| \tag{453}$$

$$v = \left|\frac{\omega\left(-\sin(r,\omega t),\cos(r,\omega t)\right)}{r}\right| \tag{454}$$

$$v = \frac{|\omega\left(-\sin(r,\omega t),\cos(r,\omega t)\right)|}{r} \tag{455}$$

$$v = \frac{|\omega|\,|(-\sin(r,\omega t),\cos(r,\omega t))|}{r} \tag{456}$$

$$v = \frac{\omega\,|(-\sin(r,\omega t),\cos(r,\omega t))|}{r} \tag{457}$$

$$v = \frac{\omega\sqrt{(-\sin(r,\omega t))^2 + \cos^2(r,\omega t)}}{r} \tag{458}$$

$$v = \frac{\omega\sqrt{\sin^2(r,\omega t) + \cos^2(r,\omega t)}}{r} \tag{459}$$

$$v = \frac{\omega\sqrt{r^2}}{r} \tag{460}$$

$$v = \frac{\omega\,|r|}{r} \tag{461}$$

$$v = \frac{\omega r}{r} \tag{462}$$

$$v = \omega \tag{463}$$

$$E = \frac{mv^2}{2} \tag{464}$$

$$E = \frac{m\omega^2}{2} \tag{465}$$

$$\vec{a}(t) = \frac{d^2\vec{r}}{dt^2} \tag{466}$$

$$\vec{a}(t) = \frac{d^2(x,y)}{dt^2} \tag{467}$$

$$\vec{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}\right) \tag{468}$$

$$\frac{d^2x}{dt^2} = \frac{d\left(\frac{\partial\cos(r,s)}{\partial s}\Big|_{s=\omega t}\frac{d(\omega t)}{dt}\right)}{dt} \tag{469}$$

$$\frac{d^2x}{dt^2} = \left(\frac{\partial^2\cos(r,s)}{\partial s^2}\cdot\frac{ds}{dt}\Big|_{s=\omega t}\right)\cdot\frac{d(\omega t)}{dt} + \frac{\partial\cos(r,s)}{\partial s}\Big|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{470}$$

$$\frac{d^2x}{dt^2} = \left(\frac{-\cos(r,\omega t)}{r^2}\cdot\frac{d(\omega t)}{dt}\right)\cdot\frac{d(\omega t)}{dt} + \frac{\partial\cos(r,s)}{\partial s}\Big|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{471}$$

$$\frac{d^2x}{dt^2} = \frac{-\cos(r,\omega t)}{r^2}\cdot\left(\frac{d(\omega t)}{dt}\right)^2 + \frac{\partial\cos(r,s)}{\partial s}\Big|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{472}$$

$$\frac{d^2x}{dt^2} = \frac{-\cos(r,\omega t)}{r^2}\cdot\omega^2 + \left.\frac{\partial\cos(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{473}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \left.\frac{\partial\cos(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{474}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r}\cdot\frac{d^2(\omega t)}{dt^2} \tag{475}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r}\cdot 0 \tag{476}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + 0 \tag{477}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} \tag{478}$$

$$\frac{d^2y}{dt^2} = \frac{d\left(\left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\frac{d(\omega t)}{dt}\right)}{dt} \tag{479}$$

$$\frac{d^2y}{dt^2} = \left(\frac{\partial^2\sin(r,s)}{\partial s^2}\cdot\left.\frac{ds}{dt}\right|_{s=\omega t}\right)\cdot\frac{d(\omega t)}{dt} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{480}$$

$$\frac{d^2y}{dt^2} = \left(\frac{-\sin(r,\omega t)}{r^2}\cdot\frac{d(\omega t)}{dt}\right)\cdot\frac{d(\omega t)}{dt} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{481}$$

$$\frac{d^2y}{dt^2} = \frac{-\sin(r,\omega t)}{r^2}\cdot\left(\frac{d(\omega t)}{dt}\right)^2 + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{482}$$

$$\frac{d^2y}{dt^2} = \frac{-\sin(r,\omega t)}{r^2}\cdot\omega^2 + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{483}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\cdot\frac{d^2(\omega t)}{dt^2} \tag{484}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \frac{\cos(r,\omega t)}{r}\cdot\frac{d^2(\omega t)}{dt^2} \tag{485}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \frac{\cos(r,\omega t)}{r}\cdot 0 \tag{486}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + 0 \tag{487}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} \tag{488}$$

$$\vec{a}(t) = \left(\frac{-\omega^2\cos(r,\omega t)}{r^2}, \frac{-\omega^2\sin(r,\omega t)}{r^2}\right) \tag{489}$$

$$\vec{a}(t) = \frac{-\omega^2\left(\cos(r,\omega t), \sin(r,\omega t)\right)}{r^2} \tag{490}$$

$$\vec{a} = \frac{-\omega^2\vec{r}}{r^2} \tag{491}$$

$$a = |\vec{a}| \tag{492}$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right| \tag{493}$$

$$a = \frac{\left|-\omega^2 \vec{r}\right|}{r^2} \tag{494}$$

$$a = \frac{\left|-\omega^2\right| \left|\vec{r}\right|}{r^2} \tag{495}$$

$$a = \frac{\omega^2 \left|\vec{r}\right|}{r^2} \tag{496}$$

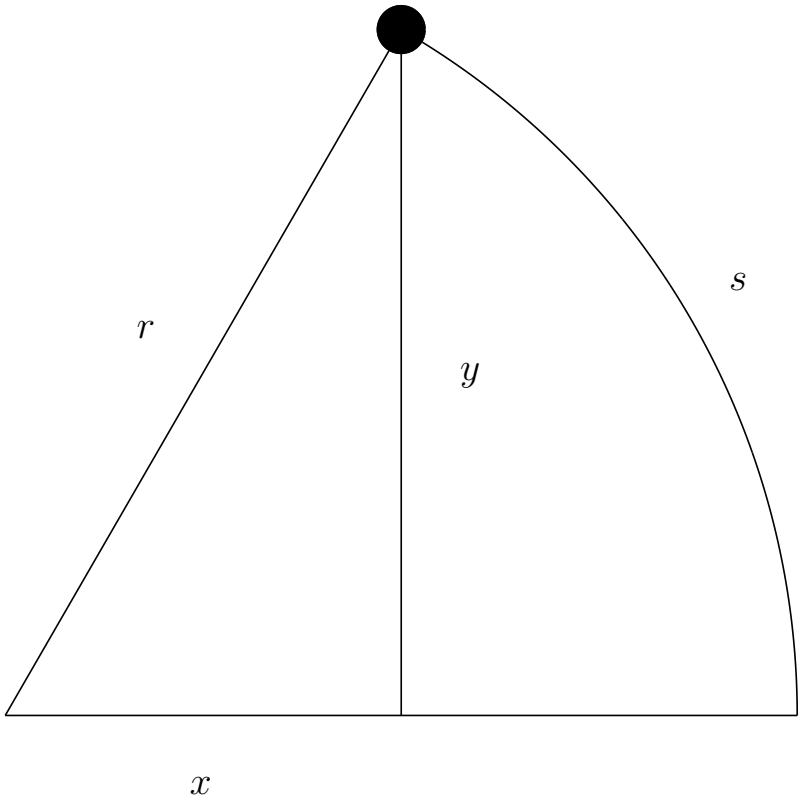
$$a = \frac{\omega^2 r}{r^2} \tag{497}$$

$$a = \frac{\omega^2}{r} \tag{498}$$

$$F = ma \tag{499}$$

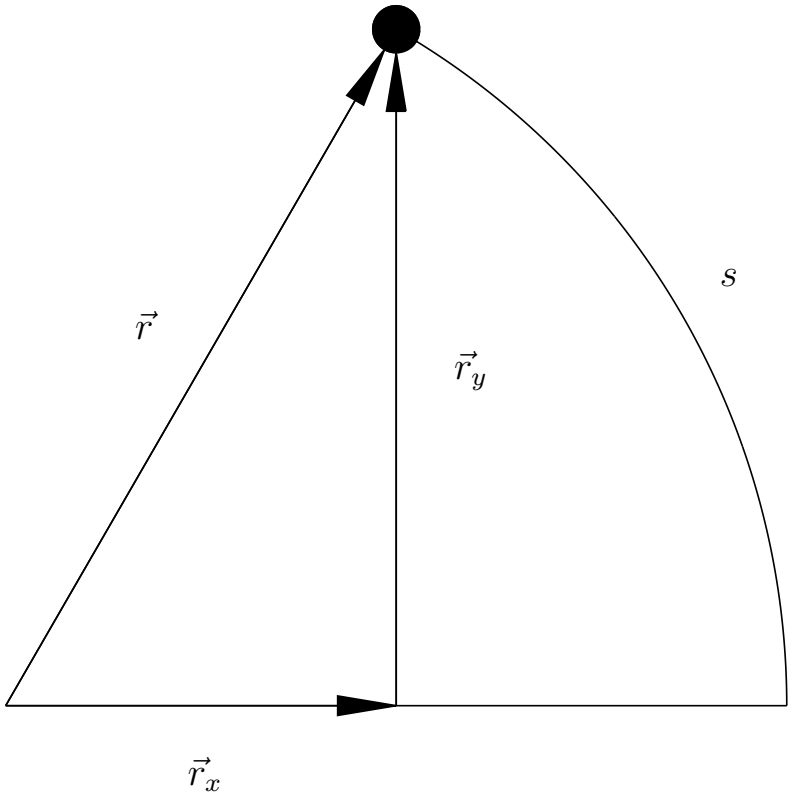
$$F = m\frac{\omega^2}{r} \tag{500}$$

$$F = \frac{m\omega^2}{r} \tag{501}$$



(a) Spatial position diagram.

Figure 5: Uniform circular motion.



(b) Position vector diagram.

Figure 5: Uniform circular motion.